(i) Course Title: ABE 721: Agricultural Systems Modelling and Simulation

(ii) Course Aim:

To impart students with knowledge and skills on developing agricultural systems and simulations

(iii) Course Expected Learning Outcomes

By the end of this course students should be able to:

1. Formulate a real-world problem and select an appropriate analytical technique for modelling and ultimately solving this problem
2. Use simulation software for model development and analysis of agricultural systems
3. Design and run simulation models in practical situations
4. Communicate the results of the modelling process to management and other non-specialist users of engineering analysis

(iv) Course Status: Core

(v) Credit Rating: 12

(vi) Total Hours Spent: 120 Hours

(vii) Course Content:

Module 1: Advanced Computer Aided Design and Analysis

1.1 Geometric modelling techniques
1.2 Systems analysis basics
1.3 Application to farm machinery scheduling problem
1.4 Design of farm machinery with the help of CAD

Module 2: Models of Systems:

2.1.Introduction and terminologies
2.2.Use of scientific models, classification of models
2.3.Modelling process; Model formulation

Module 3: Qualitative and Quantitative

3.1 Simulation paradigm
3.2 Numerical techniques
3.3 Parameter estimation

Module 4: Model Validation and Analysis

4.1 Stochastic models. Stocks and flows
4.2 Equilibrium diagrams
4.3 S-shaped growth
4.4 Causal loop diagrams
4.5 Water flows in the mono basin

Module 5: Simulating Materials Flow

1.1 Simulating Cyclical systems
1.2 Spatial and temporal models integration

1.3 Evaporation models
1.4 Crop growth and management simulation.

Practical Skills: Construct mathematical models for more complex problems.

(viii) Teaching and Learning Activities

The course will be delivered through a blended mode. Teaching and learning activities will include flipped learning activities, face to face lectures and synchronous lectures, multimedia lecture casts, case studies and scenarios, demonstrations, simulations, face to face, group discussions, practicals and excursions. In addition, visiting practitioners will be invited to give guest lectures.

(ix) Assessment Methods:

a) Continuous Assessment = 50%
   i. Online and face to face quizzes = 10%
   ii. Oral with written preparation tests = 10%
   iii. Class presentations and discussions = 10%
   iv. Case study analysis, field and practical reports = 10%
   v. Field, practical Projects or Laboratory Work performance = 10%
b) Final Assessment = 50%
   i. Complete a real-world models setup and results presentation = 25%
   ii. Project/Case Study (reports, presentations and discussions) = 25%

(x) Reading List:

1. Ahuja, L., Ma, L. and Howell, T.A. (2002). *Agricultural systems models in field research and technology transfer*. Lewis Publishers, New York, NY.
2. Ford, A. (1999). *Modeling the Environment: An introduction to system dynamics models of environmental systems*. Island Press. Washington, D. C.
3. Peart, R.M. and Curry, R.B. (1998). *Agricultural Systems Modeling and Simulation*. Marcel Dekker, Inc., New York, NY.
4. Haefner, J.W. (1996). *Modeling biological systems: Principles and applications*. Chapman & Hall. New York, NY.
5. Kramer, P.J. (2012). *Water Relations of Plants*. Academic Press, Florida.
6. Michael, A.M. (2008). Irrigation: *Theory and Practice*. Vikas Publishing House Pvt Ltd, New Delhi, India.
7. Ford, A. (1999). *Modeling the Environment: An introduction to system dynamics models of environmental systems*. Island Press. Washington, D. C. 401pp.
8. Tayfur, G. (2014). *Soft Computing in Water Resources Engineering: Artificial Neural Networks, Fuzzy Logic and Genetic Algorithms*.WIT Press. 288pp.
9. Rammmurty, T. 2001. *Computer Aided Mechanical Design and Analysis*. Tata McGraw Hill, New Delhi.
10. Mukhopadhyay, M. 2000. *Matrix, Finite Element, Computer and Structural analysis*, Oxford & IBH Publishing Co. Pvt Ltd.
11. Krishnamooorty, G. 2001. *Finite Element Analysis. Theory and Programming*. Tata McGraw Hill, New Delhi.

12. Knudra, C.V. 2000. *Numerical Control and Computer Aided Manufacturing*. Tata McGraw Hill, New Delhi.
13. Shinde, G. U., & Kajale, S. R. (2012). Design optimization in rotary tillage tool system components by computer aided Engineering Analysis. *International Journal of Environmental Science and Development*, 3(3), 279.